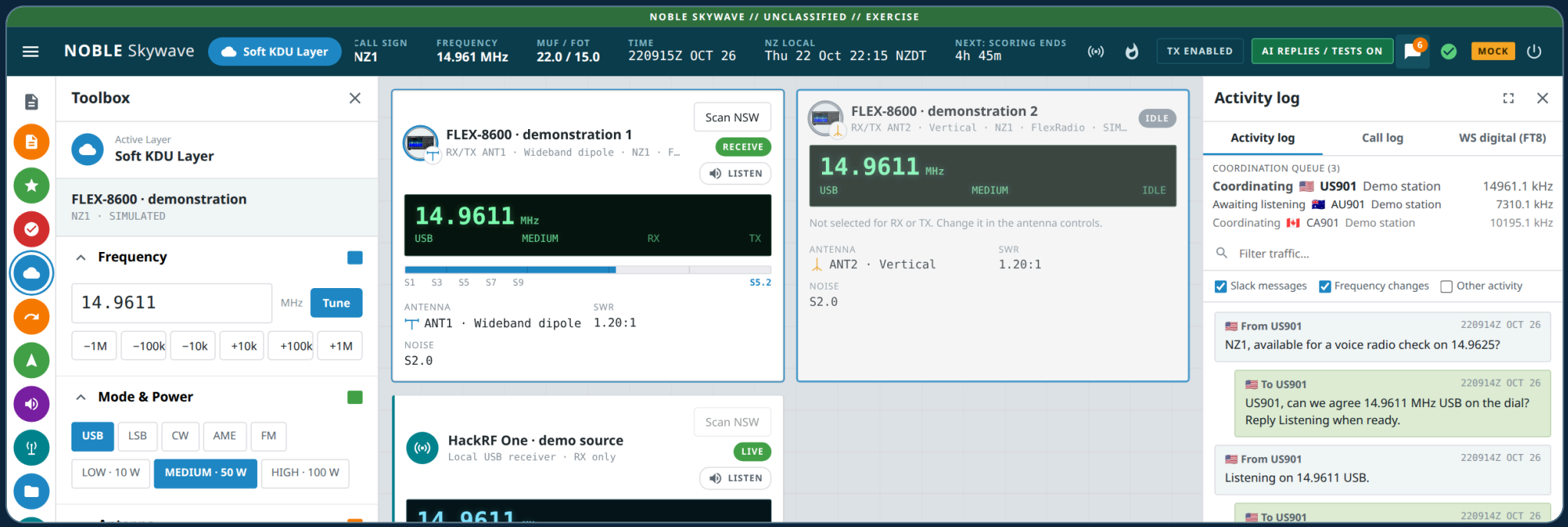
From first request to recorded contact.

A visual tour of the app that connects station coordination, guarded radio control, receive audio and reporting.



COORDINATE | TUNE & VERIFY | TRANSMIT / RECEIVE | RECORD & REPORT

SIMULATED WALKTHROUGH

One contact. One connected workflow.

The app carries the same station, dial and evidence through every stage of the exchange.

01 Discover → Find a fresh request in main chat or a peer DM.

02 Coordinate → Confirm callsign, exact dial, mode and who calls first.

03 Verify → Check the plan, actual radio, antenna, tuner and clear channel.

04 Exchange → Make one guarded call, then open the receive window.

05 Capture → Keep original audio, timing, source and receive settings.

06 Confirm → Assess the two directions using the recording and peer report.

07 Report → Prepare the contact draft and retain the delivery outcome.

Inputs

Station roster, main-chat requests, peer replies, approved channels, radio telemetry and receive audio.

Decisions

Which station is actionable, which dial is agreed, whether RX or TX is permitted, and what the evidence supports.

Outputs

Coordination messages, guarded RF checks, recordings, separate reports, local contact logs and reporting drafts.

The thread running through the app

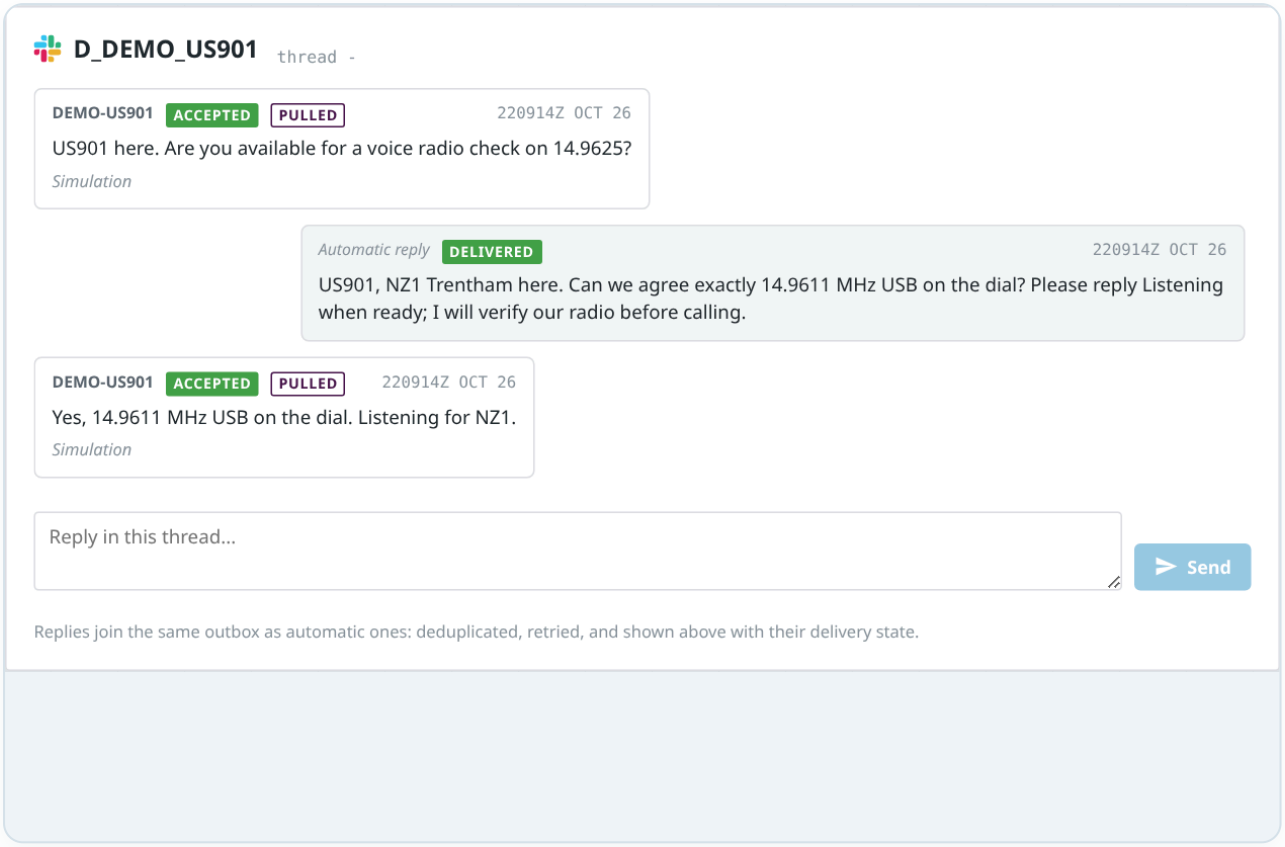
Know who is being worked, verify where the receiver really is, and keep evidence for what happened. A queue position or a sent invitation does not mean a radio call has occurred.

Demonstration scenario: NZ1 coordinates with fictional station US901 on 14.9611 MHz USB during an example exercise window. All illustrated calls, measurements, transcripts and outcomes are simulated.

SIMULATED WALKTHROUGH

01 / Find a station and agree the check.

A main-chat opportunity becomes a specific conversation with an identified station.



Actual Slack conversation view, populated offline with fictional US901 messages. Outgoing replies show their delivery state.

DISCOVERY → COORDINATION

Work fresh replies first.

1. **Match the identity**
   Compare the requester with the station roster before opening or continuing a peer exchange.

2. **Name the exact dial**
   Write the USB dial clearly in MHz and kHz. Confirm whether NZ1 should transmit or receive first.

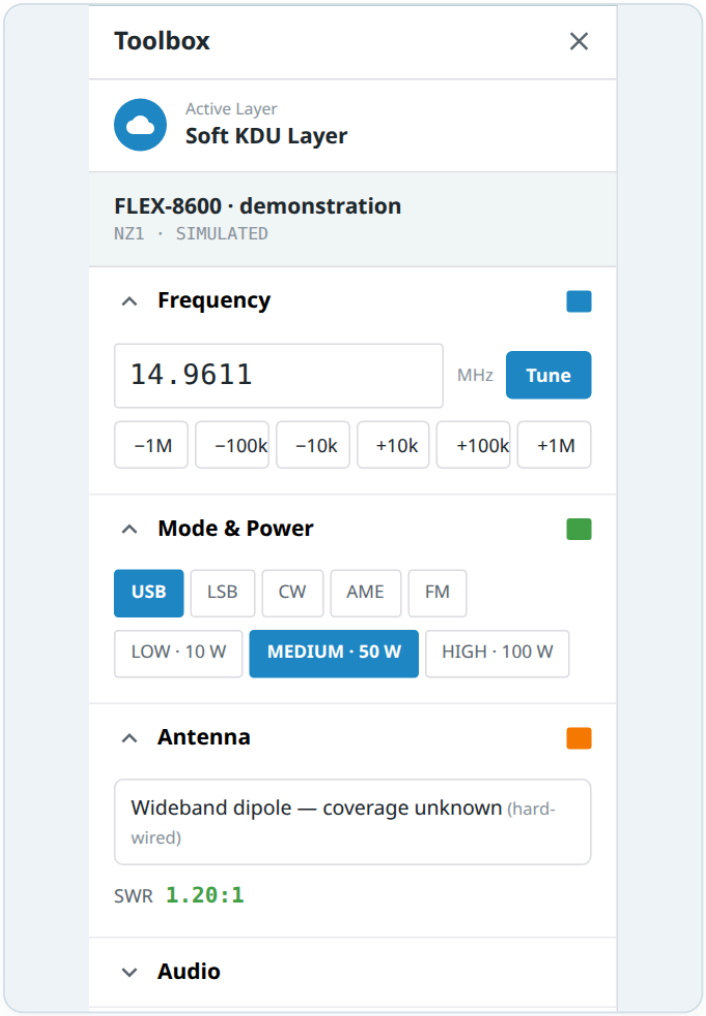
3. **Wait for readiness**
   A fresh “Listening” opens the path toward a guarded NZ1 call. If the peer is transmitting first, prepare the receiver instead.

> Several DMs can be coordinated at once. The RF exchange is serialized so stations do not compete for the same radio.

SIMULATED WALKTHROUGH

02 / Tune the radio that will do the check.

The Soft KDU connects a coordinated dial with the radio’s actual reported state.



The real Soft KDU layout: frequency, mode, power, antenna and on-air controls.

EXACT DIAL / ACTUAL READBACK

14.9611 MHz USB

Demonstration dial: 14961.1 kHz. A published channel centre and its voice dial can differ; the agreed dial must be explicit.

1. **Choose frequency and mode**
   Tune the selected source and read back the frequency and USB mode before announcing readiness.

2. **Check the selected signal path**
   Verify the receive antenna, transmit antenna and power setting. Where paired FLEX receivers are enabled, verify both are aligned.

3. **Announce what is true**
   “Listening” means the correct receiver is tuned and the receive window is actually running. A software command alone is not that proof.

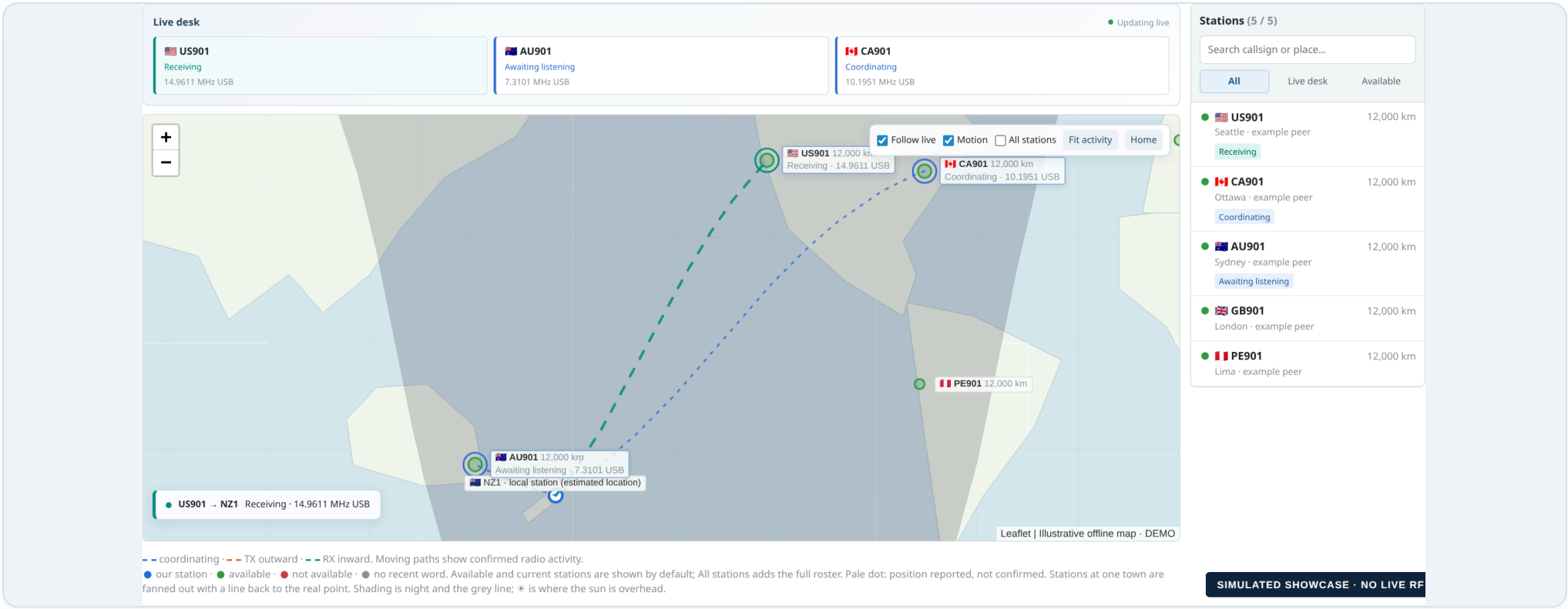
Ownership follows the radio

A live claim or manual takeover holds competing automation. The operator can inspect the same controls used by the automated desk.

SIMULATED WALKTHROUGH

03 / Use the map to see the working paths.

Live coordination and RF activity sit beside station locations and propagation planning.



Actual circuit-map UI with fictional stations and an illustrative offline basemap. The exported PDF freezes animations into one frame.

Blue / coordination

A station is being coordinated or is awaiting readiness.

Orange / transmitting

A current TX state identifies the station actually being called.

Teal / receiving

A current receive window identifies the station being monitored.

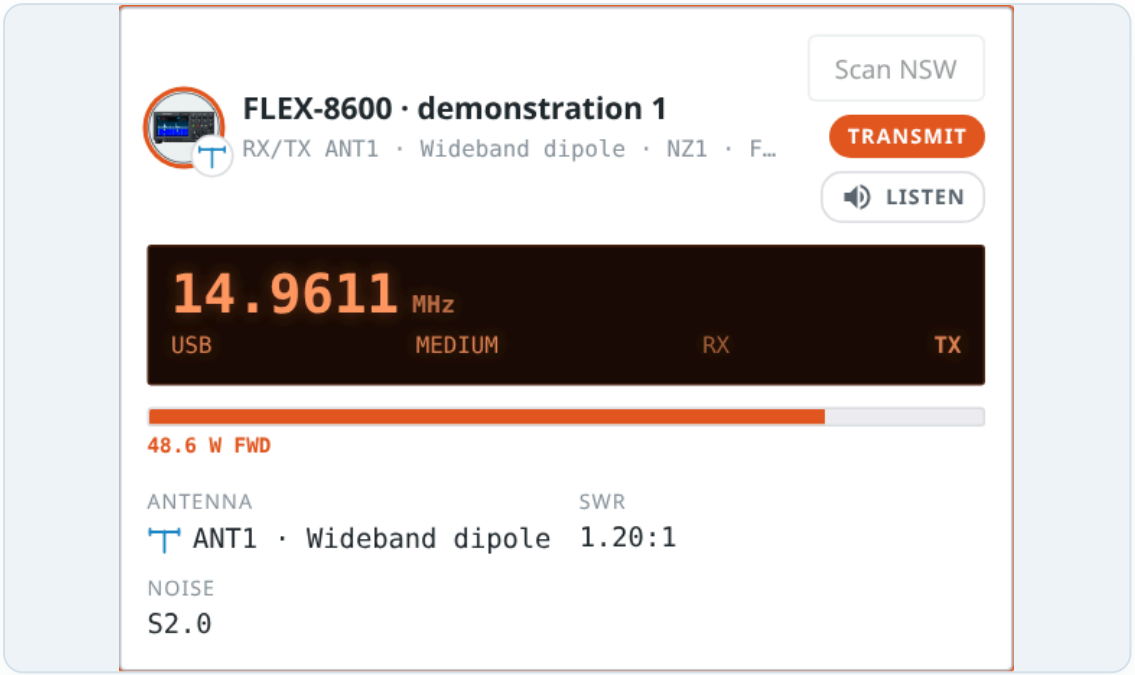
GRAFEX / planning

A forecast helps choose a path; it does not prove a contact or authorize a dial.

SIMULATED WALKTHROUGH

04 / Make one guarded voice call.

A fresh listening acknowledgement is followed by the normal preparation and clear-channel checks.



Simulated TX state in the real radio card: the agreed dial, antenna, SWR, forward power and key state are visible.

Prepare

Claim the source, verify its settings, and confirm the tuner/antenna state.

Call

Clear-channel check, then a bounded voice radio check.

Unkey

Confirm PTT released before opening the return receive window.

ONE RF PIPELINE

The call has to pass its gates.

Authorization

Approved dial, mode, call budget, station and control ownership.

Readiness

Recent peer listening reply, exact agreement and current radio readback.

RF condition

Antenna/tuner recovery, SWR, permitted power and a clear channel.

Stop conditions

EMCON, manual takeover, changed settings, duration limits or unreliable state.

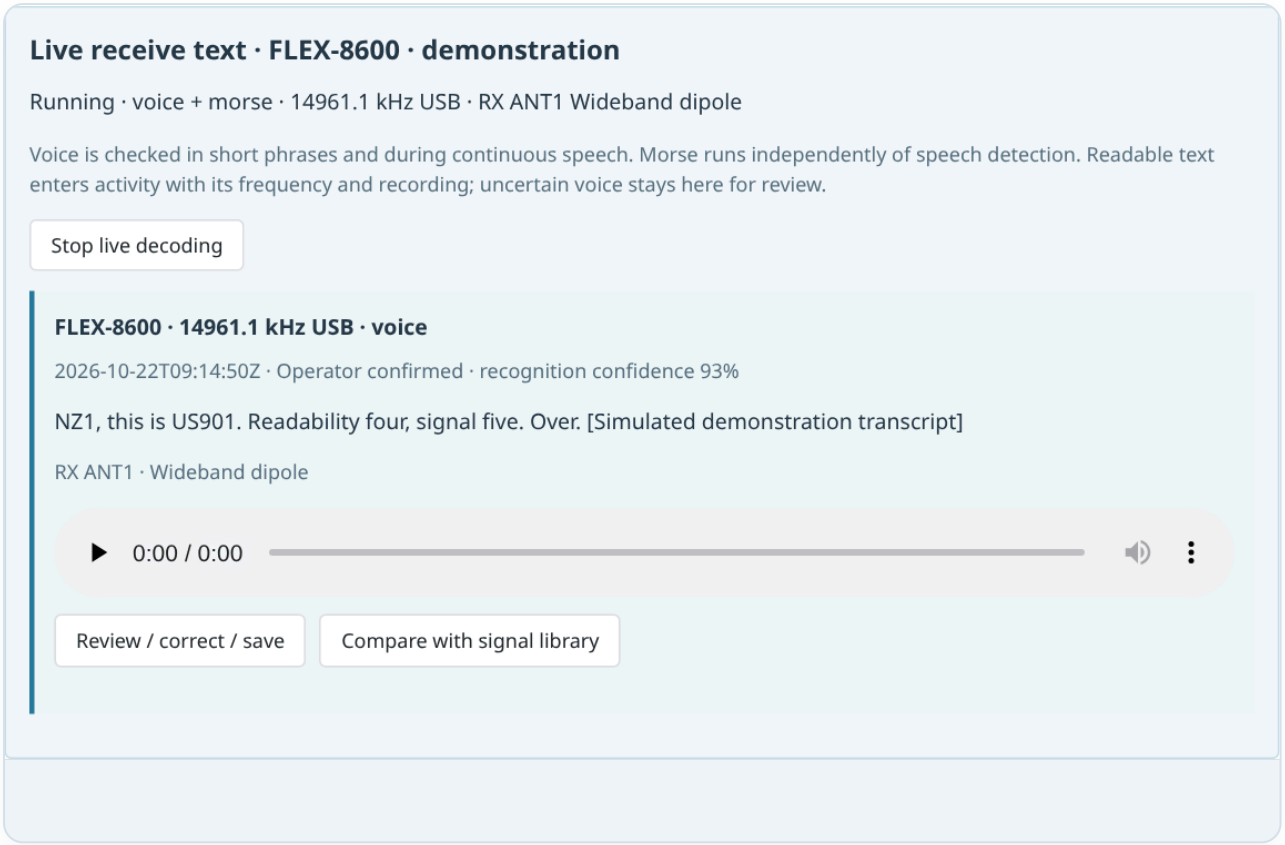
A transmission is evidence of an attempt

It becomes a successful outbound check only when the peer reports hearing NZ1. The return direction is assessed separately.

SIMULATED WALKTHROUGH

05 / Listen, decode and keep the original.

Incoming audio becomes a timed recording with its source, dial, mode and antenna context.



Actual live-receive view with an explicitly simulated US901 transcript. The app keeps decoder text and operator review distinct.

CAPTURE

Original audio

Keep the receive WAV and capture context. The original remains available for playback and download.

ASSESS

Speech and callsign

Check speech, callsign and recognition confidence. Uncertain words stay available for review.

REVIEW

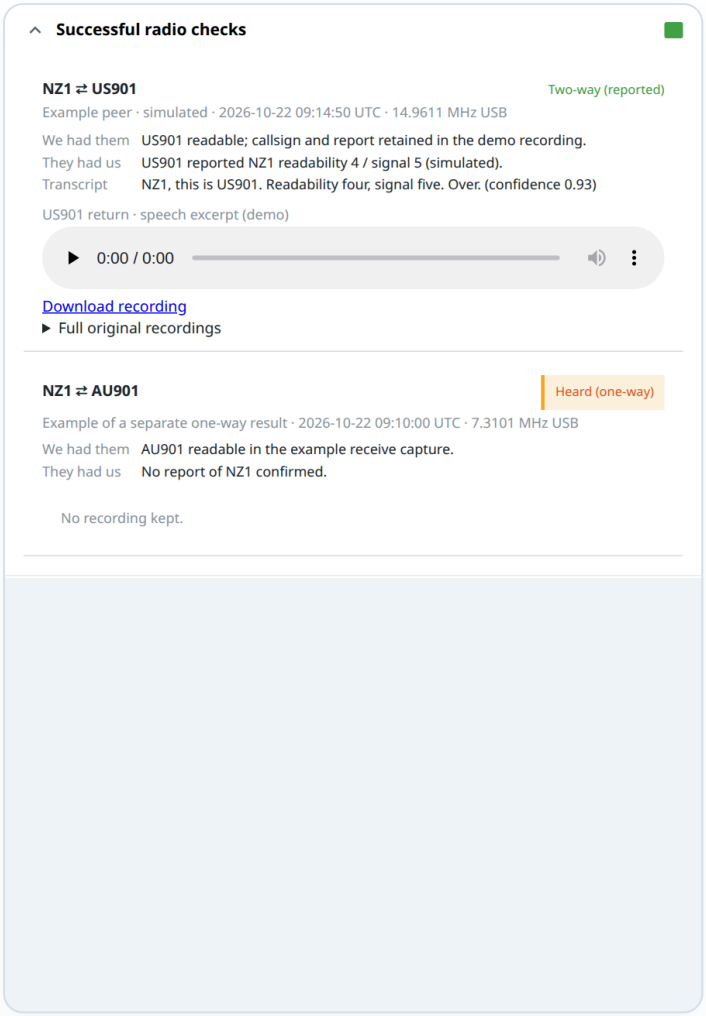
Human judgement

Listen again, correct text, confirm an observation, or keep it unresolved when the evidence is weak.

SIMULATED WALKTHROUGH

06 / Keep the two radio directions separate.

The contact log brings reports and recordings together without overstating what was confirmed.



Real contact-log component with a simulated two-way example and a separate one-way example.

EVIDENCE → RESULT

Two questions, two answers.

Did NZ1 hear the peer?

Use the recorded return, callsign and receive assessment. An operator hearing note retains its own provenance.

Did the peer hear NZ1?

Use the station's actual reception report. "Ready" and "Listening" describe readiness, not reception.

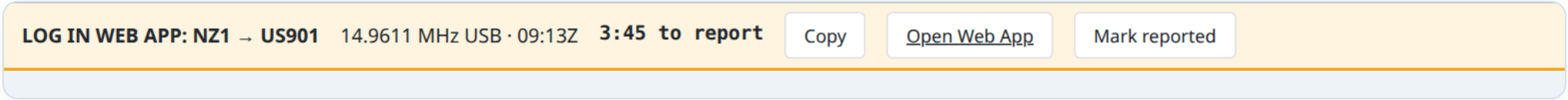
NZ1 HEARS PEER	PEER HEARS NZ1	RESULT
Yes	**Yes**	**Two-way**
Yes	No / unknown	RX only
No / unknown	Yes	TX only
Uncertain	Uncertain	**Unresolved**

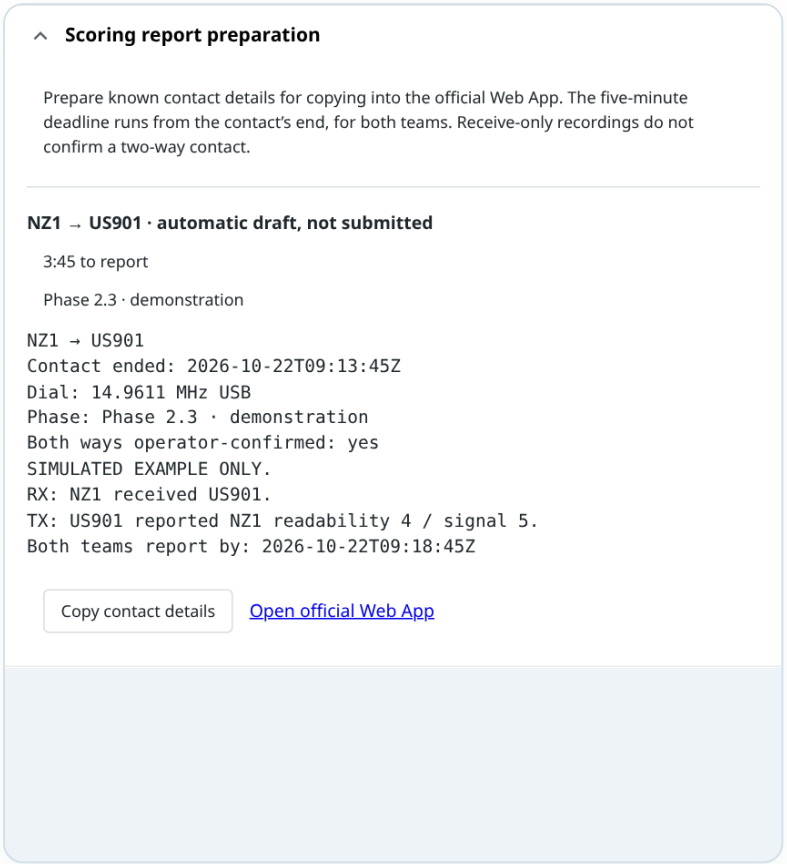
An antenna-aborted transmission remains an interrupted attempt. It does not establish a clean failed propagation result.

SIMULATED WALKTHROUGH

07 / Turn confirmed evidence into a report.

A confirmed two-way result can create a draft and a visible five-minute reporting countdown.





Simulated local reporting draft. No external contact submission or official score was created for this showcase.

1 **Prepare known details**

Carry forward callsigns, contact end time, exact dial, mode and the separate reports.

2 **Keep the deadline visible**

The clock starts at the contact's end. The banner stays visible while the operator works elsewhere in the app.

3 **Record the real submission outcome**

Copy to the official Web App, or use explicitly authorized automatic submission. Preserve the returned status and receipt.

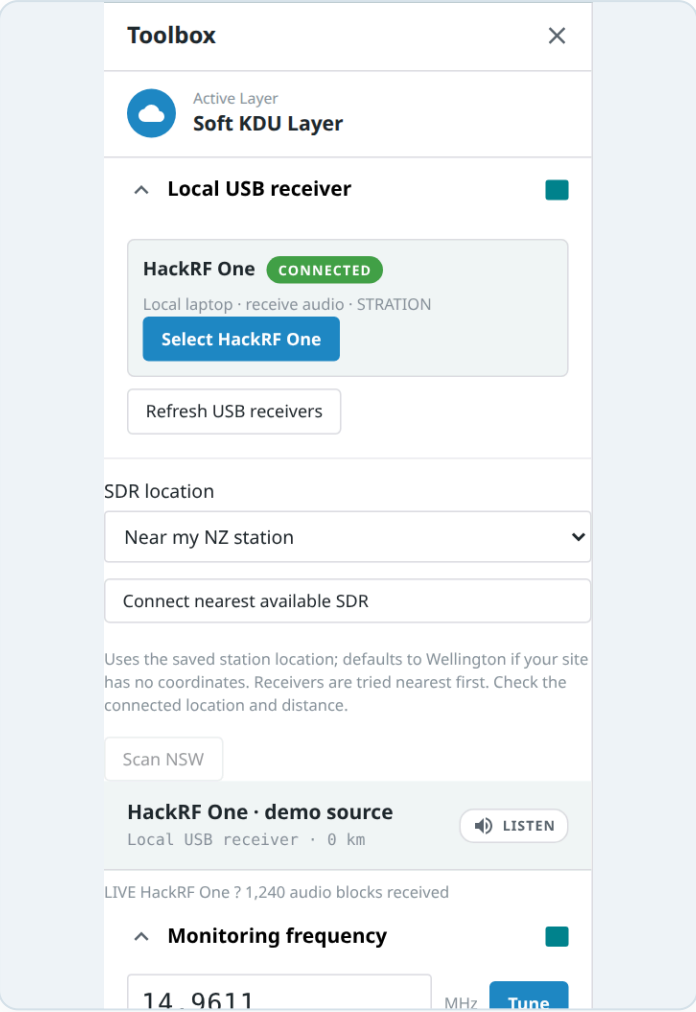
Local draft ≠ official score

"Prepared," "submitted," and the official Web App result are separate states. A receive recording alone does not create a confirmed two-way report.

SIMULATED WALKTHROUGH

Extra ears feed the same receive workflow.

The app can monitor the local radio, a local HackRF USB source or a remote KiwiSDR receiver.



Actual Soft KDU receiver controls, showing a simulated HackRF connection and audio state.

FLEX / local radio audio

The radio's receive audio enters the shared capture and speech path, alongside its telemetry and antenna context.

HackRF One / local USB

Discover and connect the USB receiver, tune its mode and dial, and select Listen. Its level is relative dBFS, not a calibrated S-meter report.

KiwiSDR / another location

Choose a public receiver near the local or distant station. Its audio describes reception at that receiver's site.

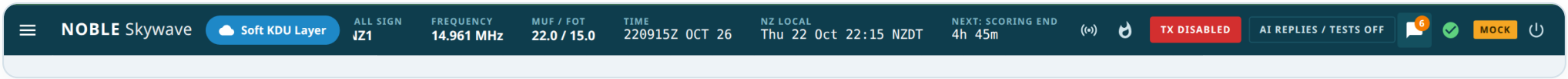
RX source → audio → recording → decode → review

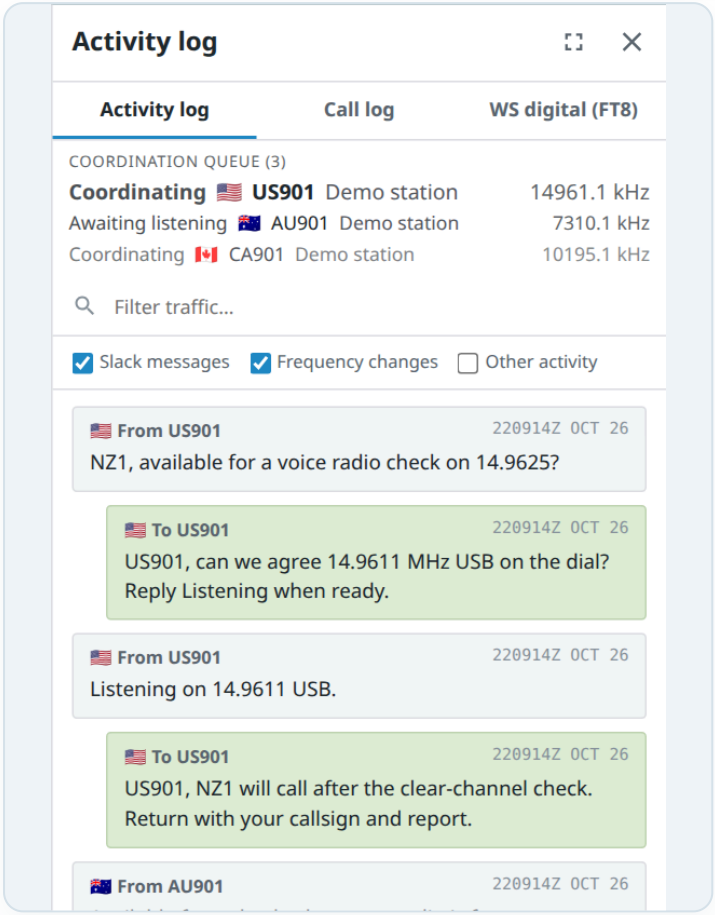
HackRF and KiwiSDR are receive-only sources in this app. A receiver may follow a radio's frequency; that does not give it transmit controls or establish what another station heard.

SIMULATED WALKTHROUGH

The operator can always see and intervene.

The activity log, queue and stop controls show the desk's current work and what still needs attention.





Actual activity log. Slack messages and frequency changes are the default filters.

1 Follow the next useful step

Calls, coordination, waiting for a listener and awaiting reports are different queue states. Eight conversations do not mean eight simultaneous transmissions.

2 Take control when needed

Manual radio changes and DM takeovers hold the affected automation. AI replies/tests and EMCON are independent controls.

3 Hand over a recoverable state

Retain actual dial, key state, active claims, remaining allowance, manual conversations, evidence and unfinished reporting work.

What a stop means

EMCON blocks RF transmission. The global automation stop and manual handoffs remain effective even when a peer is ready or a request is queued.

SIMULATED WALKTHROUGH

How the pieces connect.

A browser console brings coordination and radio evidence together through the same guarded service.

INPUTS

People and messages

Main chat, peer DMs, station roster, listening replies and reception reports.

Plans and forecasts

Approved channels, exercise phase, permitted power and path-planning information.

Radio and audio sources

FLEX telemetry and audio; HackRF USB and KiwiSDR receive streams.

APP WORKFLOW

React Soft KDU and layers

Show the same stations, controls, activity and evidence to the operator.

Guarded control service

Coordinate peers; verify readiness, claims, dial, antennas, SWR, power and stop conditions.

Evidence processing

Capture audio, decode speech or CW, support review and reconcile the two radio directions.

OUTPUTS

Station coordination and RF checks

Visible messages, bounded authorized TX, receive windows and trustworthy state changes.

Durable local evidence

Recordings, original transcripts, operator corrections, logs and contact proof packets.

Reporting and handover

Two-way drafts, deadline prompts, authorized submission results and a resumable operator state.

Request → agreement → verified radio → exchange → evidence → confirmed result → report

What this showcase demonstrates

A continuous workflow from a station asking for a check to an auditable contact result. The screenshots use the app’s built interface with isolated offline fixtures; demo peers, reports, measurements and outcomes are illustrative. No live radio was keyed, no external messages were sent, and no official scores were submitted.

Prepared from the local app and its implemented control, desk, audio, receive, logging and reporting components. Supporting screenshots and the reproducible capture/build scripts are kept beside this PDF in the project’s documentation folder.